

- 4) To make sure that Node.js is working fine, test the installation by executing the following commands:

```
$ node -v
<check for respective version>
$ npm -v
<check for respective version>
```

Once Node.js and npm are configured, we can go ahead and install Node-RED. The latest and most recent stable version of Node-RED can be downloaded from the GIT repository.

Note: To download from the GIT repo, make sure that you have configured the GIT package in your system.

```
$ sudo apt-get install git-core
$ cd ~
$ git clone https://github.com/node-red/node-red.git
$ cd node-red
$ sudo npm install
```

Now, test if Node-RED has been installed properly by executing the following command:

```
$ sudo node red.js
```

Note: If this configuration and installation is successful then we should get the following output (along with a lot of other information). We can verify this by typing `http://localhost:1880` in your favourite browser, and we should get the following output:

Node-RED available on the default 1880 port...

More details on the installation steps can be found at <http://nodered.org/docs/getting-started/installation>.

Features of Node-RED

The major features of Node-RED are listed below.

- It supports browser-based flow editing.
- As it is built on Node.js, it supports a lightweight runtime environment along with the event driven and non-blocking model.
- The various flows created in Node-RED are stored using JSON, which can be easily imported and exported for sharing with others.
- You can run it locally (Docker support, etc).
- It can easily fit on most widely used devices like Raspberry Pi, BeagleBone Black, Arduino, Android based devices, etc.
- It can run in the cloud environment like Bluemix, AWS, MS-Azure, etc.

Figure 2 explains the various components of Node-RED and its basic architecture.

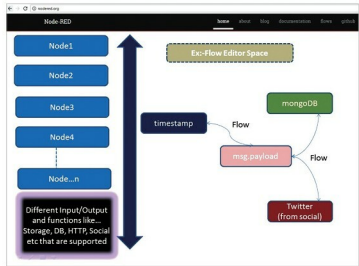


Figure 1: What a typical Node-RED flow looks like

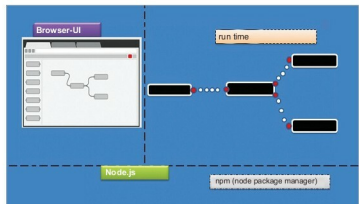


Figure 2: Architecture of Node-RED

Node-RED architecture

Some of the key aspects of Node-RED architecture are listed below:

- Node-RED (latest version 0.16) is fast because it is driven by the latest supported version of Node.js (LTS 6.x, 6.x).
- Asynchronous *io* and event-driven architecture.
- Single-threaded event queue, which supports simplicity.
- Single language support for JavaScript (both from front-end and back-end).
- Complete architecture has been built using express, d3, jquery and ws.

Different types of nodes

Inject node

- This node allows manual triggering of flows.
- It helps us to inject events at scheduled intervals.

Debug node

- The debug node helps in displaying the contents of a message—either the payload or the entire object.

Template node

- Modifies the output based on a Mustache (logic-less) template. More details can be found at <https://mustache.github.io/mustache.5.html>.

Other than the above, there are various input, output and